

Mike Silva

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Research interests: complex systems · quantum information · stochastic processes · machine learning for physics · agent-based and mean-field models · quantitative finance.

Education

Université Paris-Saclay <i>MSc Physics of Complex Systems</i> Coursework: stochastic processes, statistical field theory, disordered systems, random matrices, machine learning.	2026 – 2027 (expected) Orsay, France
Sorbonne Université <i>MSc Quantum Information</i> Coursework: quantum algorithms, quantum information theory, quantum optics, quantum metrology, ultracold atoms.	2026 – 2027 (expected) Paris, France
Université Paris Dauphine-PSL <i>MSc Economics & Finance; BSc Economics & Finance, third year (L3, 2025–2026)</i> Coursework: derivatives, portfolio management, time-series econometrics, macroeconomics.	2025 – 2027 (expected) Paris, France
Université Paris Cité <i>MSc Fundamental Physics, first year, mention Bien</i> Coursework: statistical, high-energy, computational and nonlinear physics, quantum information, condensed matter physics.	2025 – 2026 Paris, France
Sorbonne Université <i>MSc Mathematics, first year; BSc Physics and Mathematics (double major)</i> Coursework: probability, stochastic control, jump processes, functional analysis, PDEs, numerical methods, statistical physics.	2021 – 2025 Paris, France
Université Paris 1 Panthéon-Sorbonne <i>BA Economics and Political Science</i> Coursework: econometric methods, macro- and microeconomics, monetary economics, political economy.	2022 – 2025 Paris, France

Research Experience

IJCLab – Irène Joliot-Curie Laboratory (CNRS / Université Paris Cité) <i>Research Intern, supervised by Chloé Hebborn and Patrick McGlynn</i> Emulation of nuclear reaction models for uncertainty quantification on optical potentials. Thesis: <i>Computational Emulation of Nuclear Dynamics</i> , graded 18/20 (report and code).	Jun 2026 – Jul 2026 Orsay, France
- Developed a coupled-channel scattering solver based on the Direct Boundary Matching Method (DBMM), validated against an independent R-matrix calculation - Built two emulators (reduced basis method; learning reduced-order model, first coupled-channel extension) matching the solver within 5×10^{-3} (median) on four systems with up to 8 coupled channels and 10 parameters - Benchmarked both against the solver: speed-ups grow with system size, and the learning model gains 1.3–1.9× more	
CEREMADE, Université Paris Dauphine-PSL <i>Visiting Research Student, supervised by Pierre Cardaliaguet</i> Mean-field games applied to the propagation of monetary shocks in a general-equilibrium economy with sticky prices.	Sep 2024 – Jan 2025 Paris, France
- Studied differential games and mean-field models of strategic interactions between heterogeneous agents - Analyzed the coupled Hamilton–Jacobi–Bellman and Fokker–Planck equations arising in dynamic economic models - Applied mathematical and computational tools to monetary policy and collective decision-making	
Sciences Po <i>Research Assistant, supervised by Julia Cagé</i> Research assistant for <i>Hosting Media Bias</i> (Cagé et al.): built part of its dataset of French broadcasts.	Jun 2023 – Aug 2023 Paris, France

Research Projects

Urban morphogenesis: diffusion-limited aggregation and Schelling’s model <i>Université Paris Cité, supervised by Raphael Raynaud</i> Agent-based model coupling Schelling segregation with diffusion-limited aggregation: emergent morphologies and fractal dimension; extension with reinforcement learning under way.	Jan 2026 Paris, France
Self-gravitating N-body systems <i>Sorbonne Université, supervised by Lionel Foret</i> Direct-integration simulation of gravitational N-body dynamics and long-range collective effects.	Feb 2024 Paris, France

Skills

Programming	Python (NumPy, SciPy, Numba, JAX, PyTorch), R; Git, pytest, CI, LaTeX
Numerics & ML	PDE solvers, reduced-order models, Monte Carlo, neural networks, reinforcement learning
Quantum	quantum algorithms, quantum information theory, spin glasses, random matrices
Complex systems	agent-based models, mean-field games, stochastic processes, fractal growth
Economics	stochastic calculus, derivatives pricing, econometrics, general equilibrium
Languages	French (native), English (fluent), Portuguese (basic), German (basic)